

Question 1

A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation (MLE), estimate the probability of getting Heads (p).

Solution

Step 1: Understand the Problem

Maximum Likelihood Estimation (MLE) is a statistical method used to estimate the value of an unknown parameter that makes the observed data most likely. In this problem, we need to estimate the probability of getting Heads based on the observed outcomes.

For a Bernoulli or Binomial experiment, the MLE formula is:

$$\hat{p} = \text{Number of Successes} / \text{Total Number of Trials}$$

Here:

- Success = Head
- Failure = Tail

Step 2: Count the Outcomes

Outcomes:

H H T H H H T H H T

Number of Heads = 7

Number of Tails = 3

Total Tosses = 10

Step 3: Apply the Formula

$$\hat{p} = 7 / 10$$

$$\hat{p} = 0.7$$

Final Answer

The Maximum Likelihood Estimate of the probability of getting Heads is:

$$\hat{p} = 0.7$$

Interpretation

The estimated probability of obtaining a Head is **0.7 (70%)**. This means that, based on the observed sample, the coin is estimated to land on Heads 70% of the time.

Question 2

Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Solution

Step 1: Understand the Problem

Each student has only two possible outcomes:

- Pass (Success)
- Fail (Failure)

Therefore, this follows a Bernoulli distribution.

Given:

- Total students = 50
- Students passed = 42

Step 2: Apply the MLE Formula

MLE Formula:

$\hat{p}$ = Number of Successes / Total Observations

Substitute the values:

$$\hat{p} = 42 / 50$$

$$\hat{p} = 0.84$$

Final Answer

The Maximum Likelihood Estimate of the probability of passing is:

$$\hat{p} = 0.84$$

Interpretation

The estimated probability that a randomly selected student passes the examination is **0.84 or 84%**. This value best explains the observed data using Maximum Likelihood Estimation.

Question 3

A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

Solution

Step 1: Given Data

Total bulbs inspected = 100

Defective bulbs = 8

Good bulbs = 92

Step 2: Apply the MLE Formula

$\hat{p}$ = Number of Defective Bulbs / Total Bulbs

Substitute the values:

$$\hat{p} = 8 / 100$$

$$\hat{p} = 0.08$$

Final Answer

The Maximum Likelihood Estimate of the probability that a bulb is defective is:

$$\hat{p} = 0.08$$

Interpretation

The estimated probability that a bulb produced by the factory is defective is **0.08 or 8%**. This means that approximately 8 bulbs out of every 100 are expected to be defective.

Question 4

A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using Maximum Likelihood Estimation.

Solution

Step 1: Understand the Problem

Each roll has two outcomes:

- Success = Rolling a 6
- Failure = Rolling any other number

Given:

Total rolls = 60

Number of times 6 appeared = 18

Step 2: Apply the MLE Formula

$\hat{p}$ = Number of Successes / Total Trials

Substitute the values:

$$\hat{p} = 18 / 60$$

$$\hat{p} = 0.30$$

Final Answer

The Maximum Likelihood Estimate of rolling a 6 is:

$$\hat{p} = 0.30$$

Interpretation

The estimated probability of rolling the number 6 is **0.30 or 30%**.

For comparison, the probability of rolling a 6 with a fair die is:

$$1/6 = 0.167$$

Since 0.30 is much greater than 0.167, the observations indicate that the die is biased toward the number 6.

Question 5

Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

$$\text{Gradient} = 4$$

$$\text{Learning Rate} = 0.01$$

Find:

1. Predicted Output
2. Error
3. Updated Weight

Solution

Step 1: Calculate the Predicted Output

Prediction Formula:

$$y = w \times x$$

Substitute the values:

$$y = 2 \times 4$$

$$y = 8$$

$$\text{Predicted Output} = 8$$

Step 2: Calculate the Error

The error is calculated as:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Substitute the values:

$$\text{Error} = 10 - 8$$

$$\text{Error} = 2$$

$$\text{Error} = 2$$

This means the model prediction is 2 units lower than the actual output.

Step 3: Update the Weight Using Gradient Descent

Gradient Descent Formula:

$$\text{New Weight} = \text{Old Weight} - (\text{Learning Rate} \times \text{Gradient})$$

Substitute the values:

$$\text{New Weight} = 2 - (0.01 \times 4)$$

$$\text{New Weight} = 2 - 0.04$$

$$\text{New Weight} = 1.96$$

$$\text{Updated Weight} = 1.96$$

Interpretation

Since the gradient is positive, the current weight increases the prediction error. Gradient Descent reduces the weight from **2.00 to 1.96**, helping minimize the prediction error. Repeating this process many times enables the neural network to learn the optimal weight values.